Lecturers: Luke Bennetts and Chris Baker
School of Mathematics and Statistics

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University of Melbourne

MAST20030: Differential Equations
Tutorial Solutions: Week 7

Solution to Question 1.

(a) Firstly, all eigenfunctions satisfy the boundary conditions, i.e.,

$$\begin{aligned} y_0(-\pi) &= y_0(\pi) = 1/\sqrt{2}, \\ y_0'(-\pi) &= y_0'(\pi) = 0, \\ y_n^{(1)}(-\pi) &= y_n^{(1)}(\pi) = (-1)^n, \\ y_n'^{(1)}(-\pi) &= y_n'^{(1)}(\pi) = 0, \\ y_n^{(2)}(-\pi) &= y_n^{(2)}(\pi) = 0, \\ y_n'^{(2)}(-\pi) &= y_n'^{(2)}(\pi) = n(-1)^n. \end{aligned}$$

Next we check the differential operator,

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_0 = 0$$

therefore $\lambda = 0$ is the eigenvalue for eigenfunction y_0. Similarly

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_n^{(2)} = n^2 \cos nx = n^2 y_n^{(1)} \qquad \text{and} \qquad -\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_n^{(2)} = n^2 \sin nx = n^2 y_n^{(2)}$$

and so $\lambda_n = n^2$ are the eigenvalues for $y_n^{(1)}$ and $y_n^{(2)}$.

(b) We first calculate the inner product $\langle y_m^{(2)}, y_n^{(2)} \rangle$ where $m \neq n$:

$$\langle y_m^{(2)}, y_n^{(2)} \rangle = \alpha \int_{-\pi}^{\pi} \sin mx \sin nx \, \mathrm{d}x = \frac{\alpha}{2} \int_{-\pi}^{\pi} \cos(m-n)x - \cos(m+n)x \, \mathrm{d}x = 0$$

because $m - n$ and $m + n$ are always non-zero integers. Now

$$\langle y_n^{(2)}, y_n^{(2)} \rangle = \alpha \int_{-\pi}^{\pi} \sin^2 nx \, \mathrm{d}x = \frac{\alpha}{2} \int_{-\pi}^{\pi} 1 - \cos 2nx \, \mathrm{d}x = \frac{\alpha}{2} \left[x\right]_{-\pi}^{\pi} = \alpha\pi$$

and so the eigenfunctions are orthogonal. To make the eigenfunctions orthonormal we set $\alpha = 1/\pi$.

Solution to Question 2.

(a) We can write $\sin^2 x = \frac{1}{2}(1 - \cos(2x))$, so we should suspect that the Fourier series of f is just $\frac{1}{2} - \frac{1}{2}\cos(2x)$. To check this, note that f is an even function, so the coefficient of each $\sin(nx)$ vanishes. The Fourier series therefore takes the form

$$(\mathcal{F}f)(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(nx),$$

where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, \mathrm{d}x = \frac{1}{\pi} \int_{-\pi}^{\pi} \left[\frac{1}{2} - \frac{1}{2}\cos(2x)\right] \mathrm{d}x = \frac{1}{\pi} \cdot 2\pi \cdot \frac{1}{2} = 1$$

(because $\cos(2x)$ is orthogonal to 1)

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{2}-\frac{1}{2}\cos(2x)\right]\cos(nx)\,\mathrm{d}x = -\frac{1}{2\pi}\int_{-\pi}^{\pi}\cos(2x)\cos(nx)\,\mathrm{d}x$$

(because $\cos(nx)$ is orthogonal to 1, for $n \neq 0$)

$$= -\frac{1}{2\pi}\delta_{n,2}\int_{-\pi}^{\pi}\cos^2(2x)\,\mathrm{d}x$$

(because $\cos(nx)$ is orthogonal to $\cos(2x)$, if $n \neq 2$)

$$= -\frac{1}{2\pi}\delta_{n,2}\int_{-\pi}^{\pi}\left[\frac{1}{2}+\frac{1}{2}\cos(4x)\right]\mathrm{d}x = -\frac{1}{2}\delta_{n,2}$$

(because $\cos(4x)$ is orthogonal to 1). So, indeed, the Fourier series of f is $\mathcal{F}f = \frac{1}{2}-\frac{1}{2}\cos(2x)$.

(b) Again, g is an even function, so the coefficient of each $\sin(\pi nx)$ vanishes. We therefore compute

$$a_0 = \int_{-1}^{1} g(x)\,\mathrm{d}x = \int_{-1}^{1} x\,\mathrm{d}x = \frac{1}{2}x^2\Big|_{-1}^{1} = 1,$$

$$a_n = \int_0^1 g(x)\cos(\pi nx)\,\mathrm{d}x = 2\int_0^1 x\cos(\pi nx)\,\mathrm{d}x = 2\,\frac{x\sin(\pi nx)}{\pi n}\Big|_0^1 - 2\int_0^1 \frac{\sin(\pi nx)}{\pi n}\,\mathrm{d}x$$

$$= 2\,\frac{\cos(\pi nx)}{\pi^2 n^2}\Big|_0^1 = \frac{2((-1)^n-1)}{\pi^2 n^2} = \begin{cases} -\dfrac{4}{\pi^2 n^2} & \text{if } n>0 \text{ is odd,} \\ 0 & \text{if } n>0 \text{ is even.}\end{cases}$$

The Fourier series of g is therefore

$$(\mathcal{F}f)(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n\cos(\pi nx) = \frac{1}{2} - \frac{4}{\pi^2}\sum_{m=1}^{\infty}\frac{\cos(\pi(2m-1)x)}{(2m-1)^2}.$$

As this converges to the periodic extension of g and $g(0)=0$, we deduce that

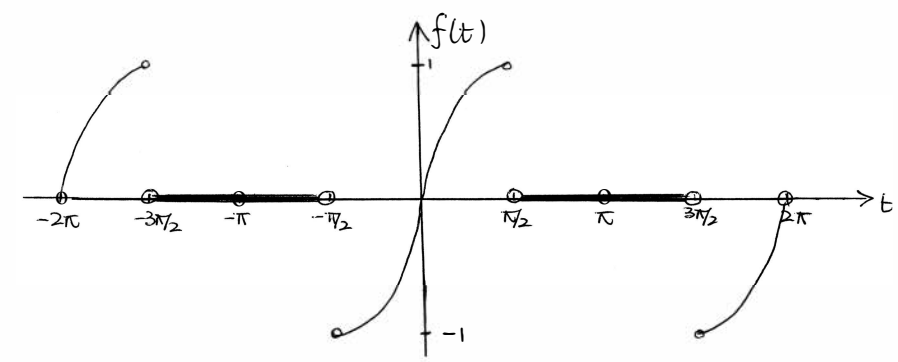
$$\frac{1}{2} - \frac{4}{\pi^2}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = 0 \;\Rightarrow\; \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{\pi^2}{8}.$$

For the last result, note that

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \sum_{m=1}^{\infty}\frac{1}{(2m)^2} + \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} + \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} \;\Rightarrow\; \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{4}{3}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{\pi^2}{6}.$$

Solution to Question 3.

(a) The function looks like



(b) By the 'pointwise convergence of a Fourier series' theorem, the Fourier series will converge to the midpoint at the discontinuity at $t = \pi/2$, i.e., the Fourier series will converge to $1/2$.

(c) Both the constant and cos terms will be zero because the integrands for the coefficients is odd. The coefficients for the sin terms are

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t) = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin t \sin nt \,\mathrm{d}t = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(n-1)t - \cos(n+1)t \,\mathrm{d}t.$$

Now for $n = 1$,

$$b_1 = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 - \cos 2t \,\mathrm{d}t = \frac{1}{2\pi}\left[t + \frac{1}{2}\sin 2t\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2}.$$

For $n \neq 1$,

$$\begin{aligned} b_n &= \frac{1}{2\pi}\left[\frac{1}{n-1}\sin(n-1)t - \frac{1}{n+1}\sin(n+1)t\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= \frac{1}{\pi}\left(\frac{1}{n-1}\sin(n-1)\frac{\pi}{2} - \frac{1}{n+1}\sin(n+1)\frac{\pi}{2}\right). \end{aligned}$$

Now $b_{2n-1} = 0$ and

$$b_{2n} = \frac{1}{\pi}\left(\frac{1}{2n-1}(-1)^{n+1} - \frac{1}{2n+1}(-1)^n\right) = \frac{(-1)^{n+1}}{\pi}\left(\frac{1}{2n-1} + \frac{1}{2n+1}\right) = \frac{4n(-1)^{n+1}}{(4n^2-1)\pi}$$

The Fourier series is therefore given by

$$(\mathcal{F}f)(t) = \frac{1}{2}\sin t + \sum_{n=1}^{\infty} \frac{4n(-1)^{n+1}}{(4n^2-1)\pi}\sin 2nt$$

(d) At $t = \dfrac{\pi}{2}$, $\sin 2nt = 0$ and $\sin t = 1$ and so $(\mathcal{F}f)\left(\dfrac{\pi}{2}\right) = \dfrac{1}{2}$.

(e) The Fourier coefficients decay as $\dfrac{1}{n}$ for large n, and so the convergence will be slow. Because the periodic extension of the function contains discontinuities at $\dfrac{\pi}{2} + m\pi$ for $m \in \mathcal{Z}$ the Fourier series does not converge uniformly and will exhibit Gibbs phenomenon.